

Using random walks in exchange rate modeling

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Abstract

Modeling and forecasting exchange rates has long been one of the most difficult challenges in international finance because exchange rates are highly volatile, influenced by a wide array of global economic, political, and financial factors. Traditional forecasting models often struggle to account for the complexity and unpredictability of exchange rate dynamics. Against this backdrop, the random walk model has emerged as a widely used benchmark for evaluating the performance of more sophisticated forecasting techniques. The random walk hypothesis assumes that changes in exchange rates are purely stochastic and not systematically predictable. In other words, today's exchange rate contains all available information, and future movements depend exclusively on the arrival of new, unforeseen information. This assumption is closely tied to the Efficient Market Hypothesis (EMH), which posits that asset prices fully reflect all publicly available information, leaving no room for consistent arbitrage opportunities. If markets are efficient, past data and observable trends cannot systematically improve forecasts of exchange rate movements.

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Chapter 1

Introduction

Modeling and forecasting exchange rates remain a fundamental yet challenging aspect of international finance due to their high volatility and complex behavior. Exchange rates are influenced by numerous unpredictable economic, political, and market factors, making reliable prediction difficult. Among various approaches, the random walk model has emerged as a key benchmark for exchange rate forecasting, grounded in the Efficient Market Hypothesis, which suggests that future exchange rate changes are independent of past movements and driven solely by new information. This study explores the effectiveness of the random walk model using historical data from major currency pairs, comparing its forecasting performance with traditional time series models. The analysis aims to provide insight into the practical utility of the random walk framework and highlight the inherent limitations faced by deterministic models in capturing the stochastic dynamics of foreign exchange rates.

1.1 Exchange Rate Modeling

Exchange rate modeling involves the development of mathematical and statistical frameworks to describe and predict the movements of currency values relative to one another. Exchange rates are influenced by a complex interplay of economic fundamentals, market sentiment, geopolitical events, and speculative activities, making their behavior highly volatile and often difficult to forecast. Traditional approaches to modeling exchange rates fall broadly into two categories: fundamental-based models and time series models. Fundamental models attempt to link exchange rate movements to macroeconomic variables such as interest rates, inflation differentials, trade balances, and monetary policies. However, these models often struggle to capture short-term fluctuations and sudden market shocks. Time series models, including autoregressive integrated moving average (ARIMA) and vector autoregression (VAR), focus on historical exchange rate data patterns to predict future values.

1.2 Random Walk Model

The random walk model is a cornerstone concept in exchange rate modeling and financial econometrics, serving as a fundamental benchmark against which the predictive performance of other models is often assessed. At its core, the random walk hypothesis posits that exchange rate changes are completely random and unpredictable, implying that the best forecast of tomorrow's exchange rate is simply today's rate plus a random error term. Mathematically, this is expressed as:

$$S_{t+1} = S_t + \varepsilon_{t+1} \quad (1.1)$$

where S_t is the exchange rate at time t , and ε_{t+1} independent, identically distributed random error term with zero mean and constant variance. The critical implication here is that past movements $S_t, S_{t-1}, \dots$ contain no useful information for predicting future rates, meaning price changes follow a “memoryless” stochastic process.

This model is deeply rooted in the Efficient Market Hypothesis (EMH), which asserts that financial markets efficiently incorporate all available information into current prices. As a result, only new, unforeseen information can cause prices to change, and because this information arrives randomly and unpredictably, exchange rates should follow a random walk. The EMH suggests that any attempt to systematically forecast exchange rate movements using historical data is inherently futile over the short term, as prices already reflect all known information.

1.3 Time series Models

In contrast to the random walk's assumption of unpredictability, time series models endeavor to exploit patterns, correlations, and structures in historical exchange rate data to improve forecast accuracy. These models are grounded in the idea that past values and errors can provide valuable information about future movements, particularly when exchange rates exhibit dependencies or volatility clustering.

One of the most widely employed classes of time series models is the Autoregressive Integrated Moving Average (ARIMA) family. ARIMA models are versatile tools designed to capture a variety of temporal patterns in time series data. In exchange rate forecasting, ARIMA models help describe short-term dependencies and smooth out noise in the data. They can identify cyclical behaviors and trends over time, making them useful for medium-term forecasting horizons. However, their performance may degrade when faced with structural breaks (e.g., major policy changes) or periods of erratic market behavior.

To specifically address the volatility dynamics often observed in financial data, Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models are frequently employed. Unlike ARIMA, which focuses on modeling the mean process, GARCH models target the variance (volatility) process. They acknowledge that volatility is not constant but clusters over time — periods of high volatility tend to be followed by high volatility, and calm periods tend to persist. This property is particularly relevant in foreign exchange markets, where geopolitical events, economic releases, or market shocks can trigger sudden spikes in volatility.

1.4 Review of the Dissertation

This dissertation provides a comprehensive examination of exchange rate modeling, focusing on the comparative performance of the random walk model and traditional time series approaches such as ARIMA and GARCH. Recognizing the inherent complexity and volatility of exchange rate dynamics, the study underscores the challenges of developing reliable forecasting models in an environment heavily influenced by unpredictable economic, political, and market factors. The random walk model is presented as a foundational benchmark, supported by the Efficient Market Hypothesis, which posits that exchange rate movements are driven by new information and are therefore inherently unpredictable. Through a detailed exploration of its theoretical underpinnings and empirical robustness, the dissertation highlights the model's strength as a baseline for evaluating more sophisticated forecasting techniques.

In contrast, the study also investigates the potential of time series models to capture underlying structure in exchange rate data. ARIMA models are shown to be effective in modeling autocorrelations and short-term dependencies, while GARCH models address the clustering behavior of volatility—a prominent feature in financial time series. Despite their methodological sophistication, these models often struggle to consistently outperform the random walk, particularly in short-term forecasting scenarios. This reinforces the notion that while statistical models can offer valuable insights into past behavior and volatility patterns, their predictive power remains limited in the face of market randomness.

Chapter 2

Literature Review

The unpredictability of exchange rates has been a central challenge in international finance and macroeconomics, with the random walk hypothesis (RWH) frequently serving as the benchmark model for exchange rate behavior and forecasting. Since the influential work of Meese and Rogoff (1983), numerous studies have tested the validity of the RWH and explored alternative approaches to improve forecast accuracy. While early research often concluded that exchange rates follow a near-random walk—making them largely unpredictable in the short run—subsequent investigations have revealed a more nuanced picture. Researchers have increasingly questioned the adequacy of linear models and the RWH’s ability to capture the complex dynamics of exchange rates, particularly over longer horizons or in the presence of nonlinear adjustments and trends.

2.1 Barbara Rossi’s (2006)

“Are Exchange Rates Really Random Walks? Some Evidence Robust to Parameter Instability” revisits the long-standing Meese–Rogoff puzzle, which claims that structural exchange-rate models based on macroeconomic fundamentals—such as interest rate differentials, inflation rates, and money supply—cannot outperform a simple random walk in forecasting future exchange rates. Since Meese and Rogoff’s influential 1983 study, this conclusion has dominated the literature, leading many to believe that exchange rates are inherently unpredictable. Rossi challenges this view by arguing that the poor performance of fundamentals may be due to parameter instability, where the relationship between exchange rates and their determinants changes over time because of shifts in policy regimes, economic structures, or external shocks. Ignoring such instability, she contends, can cause traditional forecast evaluation methods to understate the predictive power of fundamentals.

To address this issue, Rossi employs recently developed statistical tests for nested models that remain valid even when parameters are unstable. Unlike conventional approaches such as the Diebold–Mariano test, her method allows for time-varying coefficients, capturing changes in the influence of fundamentals on exchange rates over different periods.

Applying these methods to various major currencies over long historical spans, she compares structural exchange-rate models against the random walk benchmark. The findings show that once parameter instability is accounted for, the random walk is no longer universally superior. In certain cases—particularly for some countries—models based on fundamentals, especially interest rate differentials, significantly outperform the random walk both statistically and economically.

2.2 Lutz Kilian and Mark P. Taylor’s (2003)

“Why is it so difficult to beat the random walk forecast of exchange rates?” addresses the enduring challenge first highlighted by Meese and Rogoff (1983)—that models based on macroeconomic fundamentals rarely outperform the random walk in predicting exchange rates.

Rather than attributing this solely to the inadequacy of economic theory, the authors explore the possibility that traditional linear models fail because exchange rates adjust toward their long-run equilibrium in a nonlinear fashion. They focus on the exponential smooth transition autoregressive (ESTAR) model, which captures how real exchange rates mean-revert more quickly after large deviations from equilibrium and more slowly after small shocks. This framework helps resolve Rogoff’s (1996) “purchasing power parity puzzle” by explaining why real exchange rates appear to adjust glacially most of the time, while still allowing for faster adjustment in periods of substantial misalignment.

The authors develop a bootstrap-based statistical test to evaluate the random walk hypothesis for nominal exchange rates in the presence of nonlinear ESTAR dynamics. Applying this to quarterly data for seven OECD countries over the post-Bretton Woods period, they find that predictability improves markedly at longer horizons. While forecasts at short horizons remain hard to distinguish from the random walk—consistent with the idea that small deviations from fundamentals behave nearly like a random walk—models incorporating ESTAR dynamics outperform the benchmark at horizons of two to three years for six of the seven countries studied. This supports the view that the apparent unpredictability of exchange rates is partly a horizon-specific issue: predictability emerges over longer periods as nonlinear mean reversion becomes more apparent.

2.3 Francesco Lisi and Alfredo Medio’s 1997 study

“Is a random walk the best exchange rate predictor?” revisits the dominance of the random walk hypothesis (RWH) in short-term exchange rate forecasting, a result popularized by Meese and Rogoff (1983). While acknowledging the RWH as the standard benchmark, the authors argue that its apparent superiority may stem from the inability of conventional models to deal with the nonlinearities and noise inherent in exchange-rate data.

To address this, they propose a forecasting framework inspired by nonlinear dynamical systems theory, in which the raw exchange-rate series are first processed using Singular Spectrum Analysis (SSA). They adapt SSA to a nonlinear context and extend it into a multichannel version to handle multiple currency series simultaneously. This filtering step aims to extract meaningful structures from noisy data, providing a cleaner basis for forecasting.

Once filtered, the data are forecasted using a nonlinear locally linear prediction method, which is designed to capture complex, state-dependent relationships that linear models cannot detect. Applying this two-stage process to major currency exchange rates, Lisi and Medio find that their method consistently and significantly outperforms the random walk in out-of-sample, short-term forecasts. Using a test developed by Mizrach, they confirm the statistical significance of their results, reinforcing the robustness of the approach. Their findings challenge the notion that exchange rates are inherently unpredictable in the short run, suggesting instead that appropriate filtering and nonlinear modeling can reveal exploitable patterns, providing a viable alternative to the random walk benchmark in practical forecasting.

2.4 Hamid Baghestani (2009)

The study “Evaluating random walk forecasts of exchange rates” examines the performance of the random walk (RW) model as a benchmark in exchange-rate forecasting, focusing specifically on its unbiasedness and directional accuracy under symmetric loss. Using dollar/euro data for 1999–2007 and dollar/pound data for 1971–2007, the authors design a forecasting framework that generates one- to four-quarter-ahead predictions at varying lead times, enabling a direct comparison of accuracy across different horizons. They employ standard forecast evaluation techniques to test both the statistical properties of RW forecasts and their ability to correctly predict the direction of exchange-rate changes. The results indicate that forecast accuracy improves when lead times are shortened but deteriorates as the forecast horizon lengthens.

An important contribution of this study lies in its incorporation of directional accuracy into the evaluation of RW forecasts, a perspective that had not been extensively explored in earlier work. While much of the literature following Meese and Rogoff (1983) has focused on comparing RMSE or similar quantitative error measures between the RW and alternative models, this paper demonstrates that the RW’s utility extends beyond point accuracy to include reliable directional signals—at least for quarterly average rates. However, the scope is limited to average quarterly exchange rates rather than end-of-quarter rates, meaning its practical relevance is greatest for users concerned with average exchange-rate movements over a period. By showing that the RW can be both unbiased and directionally informative, the study reinforces its status as a formidable benchmark while providing a more nuanced understanding of the conditions under which it delivers useful forecasts.

2.5 G. Junge (1985)

In the study “Trends and Random Walks of Real Exchange Rates” (1985), G. Junge contributes to longstanding discussions about the properties of real exchange rates and their implications for modeling and forecasting. At the heart of the analysis lies a key empirical question: do real exchange rates follow a random walk, characterized by unpredictable, unit-root behavior, or do they exhibit deterministic trends—implying systematic dynamics that could be leveraged for forecasting? Junge’s work offers early empirical evidence that real exchange rates may not be purely random, potentially displaying underlying trends that challenge the predominant random walk hypothesis of the era. While lacking access to the full article restricts detailed summary of its methodology and findings, the title strongly suggests an investigation into distinguishing between true stochastic trends versus persistent, deterministic movements in real exchange data.

Similarly, studies exploring nonlinearity—as in ESTAR models—uncover predictability at longer horizons by allowing adjustment dynamics to vary based on the size of deviations from equilibrium. Junge’s work, emerging in the mid-1980s, may be seen as an important antecedent to such approaches—laying conceptual groundwork by identifying possible trends within real exchange rates, thereby implying that forecasts based on random walk may overlook meaningful, trend-driven components.

Chapter 3

Data Description

3.1 Data Source

The analysis was conducted using two datasets: historical USD/GBP exchange rates and USD/EUR exchange rates. Both datasets are sourced from the Federal Reserve Economic Data (FRED) platform, maintained by the Federal Reserve Bank of St. Louis. FRED aggregates and disseminates high-quality macroeconomic and financial time series data from authoritative institutions, including the Board of Governors of the Federal Reserve System. The exchange rate series are widely regarded as benchmark datasets in academic, financial, and policy analysis due to their reliability and accessibility.

1. **USD/GBP:** Daily nominal exchange rate of the U.S. Dollar relative to the British Pound Sterling.

Available at: <https://fred.stlouisfed.org/series/DEXUSUK>

2. **USD/EUR:** Daily nominal exchange rate of the U.S. Dollar relative to the Euro.

Available at: <https://fred.stlouisfed.org/series/DEXUSEU>

Both datasets reflect official market quotations from foreign exchange trading venues and are curated to ensure accuracy, continuity, and methodological transparency.

3.2 Variables

Each dataset contains two key variables:

1. **Observation Date**

- Records the calendar day of each observation.
- Format: ISO standard (YYYY-MM-DD).
- Frequency: Daily (business days only — weekends and holidays are omitted).

2. DEXUSUK / DEXUSEU

- Represents the nominal daily exchange rate of USD against GBP and EUR.
- Defined as: number of GBP (or EUR) per one USD.
- Numeric type: floating-point values, with precision to four decimal places.
- Missing values: occur on non-trading days or due to unavailable records; these must be handled during preprocessing.

3.3 Dataset Coverage

1. USD/GBP:

- Sample period: 2015-06-01 to 2025-06-30
- Raw observations: 2,610 (with 112 missing values).
- Cleaned dataset: 2,498 valid daily exchange rate entries.

2. USD/EUR:

- Sample period: 2015-06-22 to 2025-06-30
- Raw observations: 2,610 rows, of which 2,498 are valid exchange rates.
- Same structure and missing-value pattern as GBP/USD.

3.4 Data Split

For both datasets, observations were split into two subsets for forecasting and econometric modeling:

1. Training set:

- From 2015-06-01 to 2024-05-31
- Used for model fitting, trend estimation, and parameter calibration.

2. Testing set:

- 2024-06-01 to 2025-06-30
- Reserved for out-of-sample evaluation of forecasting models.

Chapter 4

Data Preprocessing

Pre-processing steps such as log transformation and differencing are not merely cosmetic adjustments for convenience; rather, they are theoretical necessities that ensure the statistical assumptions of time series models are satisfied. Financial time series, such as exchange rates, typically exhibit non-stationarity, heteroskedasticity, and multiplicative growth, which make raw data unsuitable for direct modeling.

If these issues are not addressed, models such as Random Walk, ARIMA, and GARCH will produce biased parameter estimates, invalid inference, and unreliable forecasts. Pre-processing transforms the raw series into a form with stable statistical properties (stationary mean, variance, and autocorrelation structure), making it consistent with model assumptions.

4.1 Log Transformation

Log transformation is crucial in exchange rate analysis because such series evolve multiplicatively rather than additively. Applying the natural logarithm converts these multiplicative changes into additive ones, making the data easier to model. This transformation stabilizes variance, improves interpretability by allowing log differences to be read as percentage changes (returns), and helps reduce long-term growth trends, thereby moving the series closer to stationarity, which is essential for most time series models.

4.2 Differencing Method

Why we difference a series

Many economic and financial time series are non-stationary (trends, changing mean/variance). Stationarity is required for most models (like ARIMA), and differencing is a common way to remove trends and stabilize the series.

For a series y_t , the first difference is:

$$\Delta y_t = y_t - y_{t-1}.$$

$$\Delta x_t = x_t - x_{t-1} = \ln\left(\frac{y_t}{y_{t-1}}\right) \approx \frac{y_t - y_{t-1}}{y_{t-1}}.$$

This gives log returns, which are stationary, additive, and easier to model.

Result after differencing:

The differenced log series fluctuates around a constant mean, with large spikes showing unusual shocks, making it suitable for statistical modeling.

Drift estimation:

The differenced series can be written as:

$$\Delta x_t = a + \epsilon_t$$

where a is the average per-period log return (drift) and ϵ_t is noise. It is estimated as:

$$\hat{a} = \frac{1}{T-1} \sum_{t=2}^T \Delta x_t = \frac{x_T - x_1}{T-1}$$

So the drift represents the average growth rate of the series over the period.

4.3 Stationarity Checks

1. Augmented Dickey-Fuller (ADF) Test:

To check if a time series has a unit root (i.e., non-stationary). It's a refinement of the Dickey-Fuller test, allowing for higher-order autoregressive processes.

Hypotheses:

- Null H_0 : The series has a unit root $\rightarrow$ it is non-stationary (random walk or similar).
- Alternative H_1 : The series is stationary (no unit root).

Interpretation:

- If p-value $< 0.05 \rightarrow$ reject null $\rightarrow$ the series is stationary.
- If p-value $> 0.05 \rightarrow$ fail to reject null $\rightarrow$ the series is non-stationary.

2. KPSS (Kwiatkowski–Phillips–Schmidt–Shin) Test:

To check if a time series is stationary around a mean or trend. It flips the hypotheses compared to ADF.

Hypotheses:

- Null H_0 : The series is stationary (around a mean or trend).

- Alternative H_1 : The series has a unit root $\rightarrow$ non-stationary.

Interpretation:

- If p-value $< 0.05 \rightarrow$ reject null $\rightarrow$ the series is non-stationary.
- If p-value $> 0.05 \rightarrow$ fail to reject null $\rightarrow$ the series is stationary.

4.4 Results

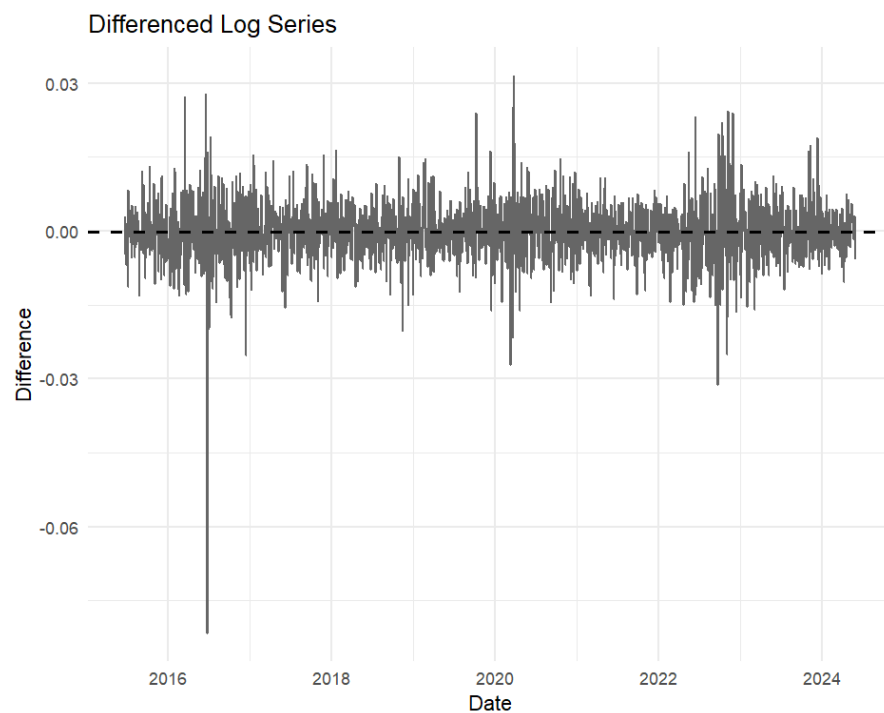


Figure 4.1: Differenced Log Series for GBP/USD Exchange Rate

Interpretation

- X-axis (Date): Runs from 2015/2016 through 2024, showing the time dimension.
- Y-axis (Difference): Represents the log differences (returns). Most values are close to 0.00, but there are spikes both positive (0.03) and negative -0.06).
- The differencing removes trends from the original log price series, making it stationary.
- What remains is the log return series (day-to-day or period-to-period percentage changes in log scale).
- The series fluctuates around zero, showing no long-term drift, but has heteroskedasticity (changing variance over time).

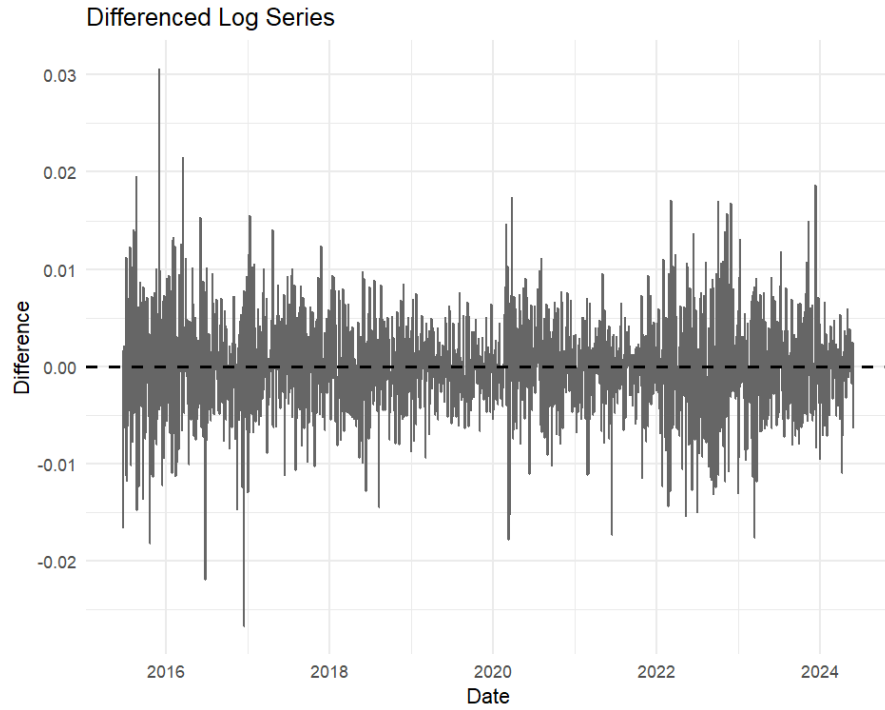


Figure 4.2: Differenced Log Series for EUR/USD Exchange Rate

Interpretation

- X-axis (Date): 2015/2016 $\rightarrow$ 2024.
- Y-axis (Difference): daily log differences, most values clustered around 0.00.
- The series oscillates around zero, showing no persistent upward or downward drift, but the variance is not constant, indicating heteroskedasticity.
- Volatility clustering is evident: periods of relative calm alternate with bursts of large fluctuations, particularly around 2016, the COVID-19 shock in early 2020, and the 2022–2023 period of global market uncertainty.
- Unlike the first differenced series, the fluctuations here are less extreme (ranging roughly between -0.02 and $+0.03$), but still reflect the impact of significant market events.

ADF and PACF Results

Table 4.1: ADF and KPSS Test Results

Currency Pair	Estimated Drift a	ADF p-value	KPSS p-value	Stationarity Conclusion
GBP/USD	-9.70792×10^{-5}	0.01	0.1	Stationary
EUR/USD	-2.14348×10^{-5}	0.01	0.1	Stationary

Summary

The differenced log series (log returns) for both USD/GBP and USD/EUR exchange rates show fluctuations around zero, confirming stationarity. The USD/GBP series exhibits higher day-to-day variability with noticeable spikes during major events, such as COVID-19 in 2020, while the USD/EUR series is generally smoother but shows occasional large negative outliers, reflecting rare extreme shocks. Both series display volatility clustering, typical of financial data.

The estimated drift (α) is very small for both series, indicating minimal average log returns per period:

- GBP/USD: -9.70792×10^{-5}
- EUR/USD: -2.14348×10^{-5}

The ADF and KPSS tests support these observations:

- ADF test: p-values of 0.01 for both series reject the null hypothesis of a unit root, indicating stationarity.
- KPSS test: p-values of 0.1 fail to reject the null hypothesis of stationarity.

Overall, both series are stationary and suitable for time series modeling.

Chapter 5

Methodology

In analyzing exchange rate dynamics, it is crucial to account for both deterministic trends and stochastic fluctuations. We compare several models commonly used in finance: the Discrete Random Walk as a benchmark, ARIMA for autocorrelation and non-stationarity, the White Noise Random Walk as a baseline with Gaussian innovations, and GARCH for time-varying volatility and clustering. Together, these models progress from simple to complex, offering insights into both mean and variance dynamics.

After defining the models, we evaluate their forecasting performance through simulation and error measurement. Using fitted parameters, we generate 1,000 forecast paths at three horizons—3, 90, and 250 days—on the log scale. Forecasts are compared with actual test data visually and quantitatively using the Root Mean Squared Error (RMSE). This framework highlights short- versus long-term accuracy, assesses robustness across datasets, and combines graphical inspection with formal metrics to identify the most reliable model.

5.1 Discrete Random Walk

5.1.1 Introduction

The discrete random walk model provides a simple yet effective framework for modeling exchange rate movements under uncertainty. It assumes that future changes are driven by random shocks, making them inherently unpredictable, consistent with the Efficient Market Hypothesis. By discretizing continuous log-returns using a scaling factor, the model transforms them into a finite set of possible increments, allowing for tractable empirical analysis. This approach captures the essential stochastic nature of exchange rates while simplifying simulation and estimation.

In the random walk model of a time series

$$X_t = D_t + W_t,$$

where D_t is the deterministic trend, and W_t is the random walk component.

$$W_t = W_{t-1} + \epsilon_t,$$

The variable ϵ_t represents the random innovation or shock at time t . It captures the unpredictable increment or decrement that drives the stochastic component W_t .

These shocks are assumed to be independent and identically distributed (i.i.d.), making them the fundamental source of randomness in the process.

5.1.2 Role of ϵ_t

- ϵ_t captures the unpredictable change in returns.

$$\epsilon_t = \begin{cases} 2 & \text{with probability } q_1, \\ 1 & \text{with probability } p_1, \\ 0 & \text{with probability } 1 - 2p - 2q, \\ -1 & \text{with probability } p, \\ -2 & \text{with probability } q_2. \end{cases}$$

- These values represent possible increments or decrements of the random walk at each step.

5.1.3 Scaling Constant K

- Since log-returns are continuous, we rescale them using:

$$X_t = K \cdot \log S_t,$$

where S_t is the exchange rate and K is a scaling constant.

- Multiplying by K and rounding transforms continuous log-returns into discrete values suitable for the random walk model.

Even if $K = 100$ already puts $\approx 95\%$ of values in $-1, 0, 1$, choosing a larger K ensures the tails are represented, the extreme states $-2, +2$ actually get used, and simulations don't underestimate volatility. It's basically trading a little central accuracy for better tail coverage and more realistic dynamics.

5.1.4 Discrete Distribution

To construct a discrete innovation distribution from the empirical log-returns, we first scaled the returns by a constant factor K and then rounded them to the nearest integer. The scaling constant K was chosen such that the vast majority of rounded increments fall within the set $\{-1, 0, 1\}$, holding more than 95% of the values, while still allowing the occasional values ± 2 . In this way, the effective support of the innovations is restricted to $\{-2, -1, 0, 1, 2\}$.

Any observations falling outside this range were handled by redistribution: their probability mass was reassigned to the core support $\{-2, \dots, 2\}$. This was done proportionally to the observed frequencies in the core bins (an alternative procedure would be to winsorize the extreme values at the boundaries -2 or 2).

To further simplify the model and remove asymmetries, we enforced symmetry of the distribution by averaging the probabilities of mirror-image bins (e.g. matching -1 with $+1$). The resulting innovation distribution is therefore parameterized as

$$\begin{aligned}P(\epsilon_t = \pm 2) &= q, \\P(\epsilon_t = \pm 1) &= p, \\P(\epsilon_t = 0) &= 1 - 2p - 2q.\end{aligned}$$

where p , q and $1 - 2p - 2q$ are estimated from the adjusted, symmetric frequencies.

Finally, we perform sanity checks to ensure that all probabilities are valid (non-negative and summing to one) and that the central mass around $\{-1, 0, 1\}$ achieves the target threshold. If these conditions were not met, the scaling factor K or the redistribution scheme was revised accordingly.

5.2 ARIMA Model

5.2.1 Introduction

An ARIMA model (AutoRegressive Integrated Moving Average) is one of the most widely used statistical models for time series forecasting. It is especially useful when data show temporal dependence (today's values are influenced by yesterday's) but do not have strong seasonal patterns.

An ARIMA model is a forecasting technique that combines autoregression (AR), integration/differencing (I), and moving average (MA) components. The AR part captures dependence on past values, the MA part captures dependence on past shocks (errors), and differencing ensures stationarity. This makes ARIMA especially effective for modeling financial and economic time series such as exchange rates, where past values and shocks influence future movements.

The Components of ARIMA

The name ARIMA(p, d, q) comes from its three main components:

1. AR – Autoregressive (p) The autoregressive part assumes that the current value of a time series depends on its past values. It is given by:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t$$

where $\phi_1, \phi_2, \dots, \phi_p$ are parameters to estimate, and ϵ_t is random noise.

2. I – Integrated (d) The integration part refers to differencing the data to make the series stationary. Stationarity means that the series has a constant mean and variance over time, which is a key requirement for ARIMA.

Differencing is applied as:

$$Y'_t = Y_t - Y_{t-1}$$

This can be repeated d times if necessary.

3. MA – Moving Average (q) The moving average part models the current value as a function of past forecast errors (shocks). It is represented as:

$$Y_t = \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $\theta_1, \theta_2, \dots, \theta_q$ are parameters to estimate.

5.2.2 Model Fit using `auto.arima()`

The `auto.arima()` function (R package `forecast`, by Rob Hyndman) automatically selects the best ARIMA model for a given time series, avoiding manual trial-and-error.

Process:

1. Stationarity check: Tests for unit roots and determines differencing order d .
2. Model exploration: Tries combinations of (p, q) and estimates parameters via Maximum Likelihood Estimation (MLE).
3. Model selection: Compares models using information criteria (AIC, AICc, BIC). By default, the model with the lowest AICc is chosen.
4. Seasonality check: If applicable, seasonal ARIMA terms (P, D, Q) are also tested.

Advantages:

- Automates model identification and tuning.
- Handles both seasonal and non-seasonal ARIMA.
- Selects statistically justified models with optimal balance of fit and complexity.

5.2.3 Auto ARIMA Model Selection

1. Data Preparation: The series was log-transformed to stabilize variance. Unit root tests suggested non-stationarity, so first differencing ($d = 1$) was applied to achieve stationarity.

2. Model Selection: `auto.arima()` systematically evaluated multiple (p, d, q) models using MLE and selected the one with the lowest AICc. With `stepwise = FALSE`, the algorithm exhaustively searched the parameter space.

3. Maximum Likelihood Estimation (MLE): MLE estimates ARIMA parameters by maximizing the likelihood of observing the given data. Log-likelihood is used for computational convenience. To avoid overfitting, model comparison relies on information criteria.

4. Information Criteria:

- **AIC:**

$$AIC = -2 \cdot \text{LogLik} + 2k$$

where k is the number of estimated parameters. Lower AIC indicates better trade-off between fit and complexity.

- **AICc:** Corrects AIC for small sample sizes, penalizing complex models more strongly. For large n , $AICc \approx AIC$.
- **BIC:** Similar to AIC but applies a stronger penalty for complexity.

5. Why AICc? By default, `auto.arima()` minimizes AICc, which provides more reliable model selection when sample size is limited.

6. Autocorrelation Function Correlation between X_t and its lagged values X_{t-k} :

$$\rho_k = \frac{\sum_{t=k+1}^T (X_t - \mu)(X_{t-k} - \mu)}{\sum_{t=1}^T (X_t - \mu)^2}, \quad -1 \leq \rho_k \leq 1, \quad \rho_0 = 1$$

Interpretation:

- $\rho_k > 0$: positive relation,

- $\rho_k < 0$: negative relation,
- $\rho_k \approx 0$: no relation.

7. Partial Autocorrelation Function: Correlation between X_t and X_{t-k} *after removing effects of lags* $1, \dots, k-1$. Estimated via $\text{AR}(k)$:

$$X_t = \alpha + \phi_{k1}X_{t-1} + \dots + \phi_{kk}X_{t-k} + \varepsilon_t$$

PACF at lag k is ϕ_{kk} .

Key: $\text{PACF}(1) = \text{ACF}(1)$; for $k > 1$, PACF shows only the direct effect.

5.3 White Noise Random Walk Model

5.3.1 Introduction

The White Noise Random Walk is a fundamental benchmark in time series modeling of exchange rates. It assumes that the *logarithm of exchange rates* evolves as a random walk, where the increments are independent and identically distributed (i.i.d.) Gaussian random variables. This simple model provides a baseline for evaluating the performance of more sophisticated forecasting models such as ARIMA.

Formally, the model can be expressed as:

$$\log(S_t) = \log(S_{t-1}) + \epsilon_t,$$

where:

- S_t is the exchange rate at time t ,
- ϵ_t are i.i.d. Gaussian shocks with mean μ (drift) and variance σ^2 , i.e.,

$$\epsilon_t \sim \mathcal{N}(\mu, \sigma^2),$$

- μ represents the average change (drift) per time step,
- σ represents the typical fluctuation magnitude (volatility).

5.3.2 Parameter Estimation

The parameters of the model are typically estimated using the *log-returns* of the observed exchange rate series:

$$r_t = \log(S_t) - \log(S_{t-1}) \sim \mathcal{N}(\mu, \sigma^2).$$

The mean of the log-returns estimates the drift μ , while the standard deviation of the log-returns estimates the volatility σ .

5.3.3 Interpretation and Insights

- A small or near-zero drift indicates that, on average, the exchange rate does not exhibit a strong directional trend over short periods.
- The volatility parameter reflects the typical short-term fluctuations of the exchange rate. Larger volatility implies more pronounced day-to-day variability.
- This model captures the empirical observation that exchange rates often behave like a near-random walk: unpredictable in the short term with small directional bias but measurable variability.

Despite its simplicity, the White Noise Random Walk model serves as a useful benchmark, allowing analysts to evaluate whether more complex models provide meaningful predictive improvements over a baseline of randomness.

5.4 GARCH Model

5.4.1 Introduction

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is widely used in financial time series analysis to capture time-varying volatility. Many financial series, such as exchange rates or stock returns, exhibit volatility clustering—periods of high variance followed by high variance, and low variance followed by low variance.

The standard GARCH(p, q) model is given by:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

- p = number of lagged conditional variances
- q = number of lagged squared innovations

5.4.2 Model Fitting

- Start simple with GARCH(1,1). It often captures most volatility dynamics.
- Check if including ARMA terms improves the mean model.
- If there is evidence of asymmetry (leverage effects), consider EGARCH or GJR-GARCH.

Use a software package (like rugarch in R) to estimate parameters via maximum likelihood. After fitting:

- Examine the estimated parameters to ensure they are positive and satisfy stationarity conditions $\alpha + \beta < 1$).
- Check robust standard errors to evaluate statistical significance.

5.4.3 Model Selection

Multiple candidate GARCH-type models can be considered:

1. Information Criteria: Akaike (AIC), Bayesian (BIC), Shibata, Hannan-Quinn.
 - Lower values indicate a better trade-off between fit and complexity.
2. Alternative models:
 - Higher-order GARCH(p,q) if volatility is more persistent.
 - EGARCH or GJR-GARCH if asymmetric responses to shocks are present.

5.4.4 Model Diagnostics

1. Ljung–Box Test

H_0 : Residuals show no serial correlation, H_1 : Residuals exhibit serial correlation.

The decision rule is based on the p -value. If the p -value is small (typically < 0.05), we reject H_0 and conclude that autocorrelation remains in the residuals. If the p -value is large (> 0.05), we fail to reject H_0 , indicating that the residuals are free from serial correlation and the mean specification is adequate.

2. ARCH LM Test

H_0 : There are no remaining ARCH effects, H_1 : There are remaining ARCH effects.

The decision again depends on the p -value. A small p -value (< 0.05) leads us to reject H_0 , suggesting that volatility clustering remains and the GARCH(1,1) specification is insufficient. A large p -value (> 0.05) means we fail to reject H_0 , implying that the conditional heteroskedasticity has been adequately captured by the model.

If the residuals are uncorrelated and show no ARCH effects, the model adequately captures the volatility structure.

Consider alternatives if:

- Residual diagnostics show remaining autocorrelation or ARCH effects.
- The series exhibits leverage/asymmetry (large negative shocks increase volatility more than positive shocks).

5.5 Model Evaluation and Forecast Simulation

After defining the models (Random Walk, White Noise Random Walk, Discrete Random Walk, and GARCH), the next step is to apply each method to the two datasets and evaluate their forecasting performance.

5.5.1 Forecast Simulation

To assess predictive ability, we simulate 1,000 paths from each model using the fitted parameters. Forecasts are generated across three horizons:

- Short-term: 3 days ahead
- Mid-term: 90 days ahead
- Long-term: 250 days ahead

All forecasts are carried out on the log scale to ensure consistency with model assumptions and to stabilize variance. For each horizon, the simulated paths are plotted alongside the actual test data, allowing for visual inspection of model fit and variability.

5.5.2 Comparison with Actual Test Data

After generating the simulated forecasts, each term of the model is compared directly with the corresponding term from the actual test dataset using the Root Mean Squared Error on the log scale (log RMSE). This ensures that the evaluation reflects relative errors between forecasted and actual values, rather than being dominated by large absolute differences. Each log RMSE is then rescaled by multiplying by 100, making it interpretable as a percentage error.

The comparisons are performed at three forecast horizons:

- 3-day horizon: Forecasted values are compared with the first 3 days of the test dataset.
- 90-day horizon: Forecasted values are compared with the first 90 days of the test dataset.
- 250-day horizon: Forecasted values are compared with the full 250-day test dataset.

This step highlights how each model performs under different time horizons, allowing evaluation of both short-term accuracy and long-term stability in relative terms.

5.5.3 Performance Metric: Root Mean Squared Error (RMSE)

To formally evaluate predictive accuracy on a relative scale, we use log RMSE, defined as:

$$\text{Log RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (\log(y_t) - \log(\hat{y}_t))^2}$$

where:

- y_t = actual observed value at time t
- $\hat{y}_t$ = predicted value at time t
- n = number of forecasted points

The log RMSE is then **rescaled** to percentage terms:

$$\text{Rescaled Log RMSE (\%)} = \text{Log RMSE} \times 100$$

Interpretation:

- Relative error magnitude: Measures average proportional deviation between forecasts and actual values.
- Squaring effect: Errors are squared to ensure positive and negative deviations do not cancel out.
- Sensitivity to outliers: Larger relative errors are penalized more strongly.
- Direct interpretability: Expressed as a percentage, indicating the average percent error in forecasts.

5.5.4 Model Comparison

By computing rescaled log RMSE for each model, each forecast horizon, and across both datasets, we establish a comprehensive evaluation framework.

This enables us to:

- Identify which model performs best for short-term, mid-term, and long-term forecasts in relative terms.
- Compare performance across datasets, testing the robustness of each model.

In this way, both visual inspection (forecast paths plotted against actual values) and quantitative evaluation (rescaled log RMSE scores) are combined, providing a balanced judgment of the suitability and reliability of each modeling approach.

Chapter 6

Results and Discussion

In this chapter, we present the results of applying the four models introduced in the methodology—Discrete Random Walk, ARIMA, White Noise Random Walk, and GARCH—to two datasets: the daily nominal exchange rate of the U.S. Dollar relative to the British Pound Sterling (USD/GBP, DEXUSUK) and the U.S. Dollar relative to the Euro (USD/EUR, DEXUSEU). The models are evaluated through forecast simulations at different horizons, and their predictive performance is compared against the actual test data using Root Mean Squared Error (RMSE). This allows for both visual and quantitative assessment of each model’s accuracy and robustness across currencies and time horizons.

6.1 Discrete Random Walk Model

6.1.1 Model Fitting for GBP/USD

1. Choice of Scaling Constant K

To discretize the innovations ϵ_t , a scaling constant K was chosen by testing a range of values. For the USD/GBP series, a final choice of $K = 125$ was made, which provided a balance between resolution (capturing finer fluctuations) and concentration (ensuring most of the probability mass was contained within the support).

2. Raw Innovation Distribution ϵ_t

For USD/GBP, most of the scaled innovation values ϵ_t fell between -2 and $+2$, with very little probability mass lying outside this range (e.g., occasional values such as -3 , $+3$, or $+4$). Since such extreme movements were rare, the distribution was truncated to the support $\{-2, -1, 0, 1, 2\}$ and all probability mass outside this range was proportionally redistributed into the tails.

3. Final Discrete Probability Distribution

After redistributing the mass, the final estimated probabilities of ϵ_t for USD/GBP were:

ϵ_t	Probability (%)
-2	2.03
-1	18.80
0	58.34
1	18.80
2	2.03

Table 6.1: Discrete probability distribution of ϵ_t for GBP/USD

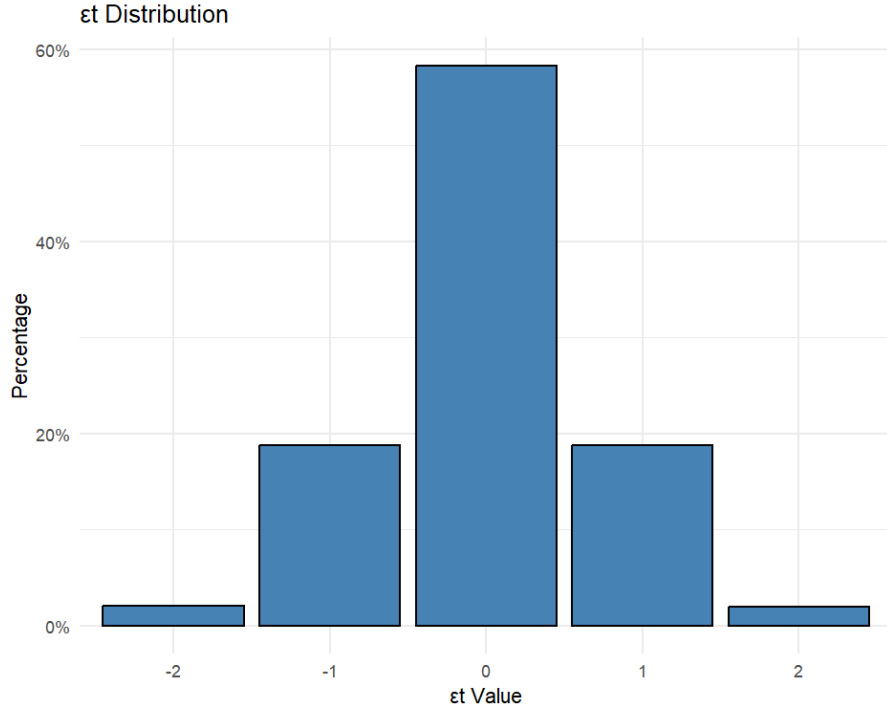
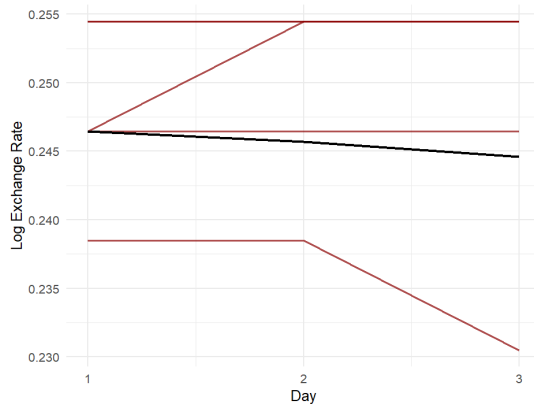


Figure 6.1: Distribution Plot for GBP/USD

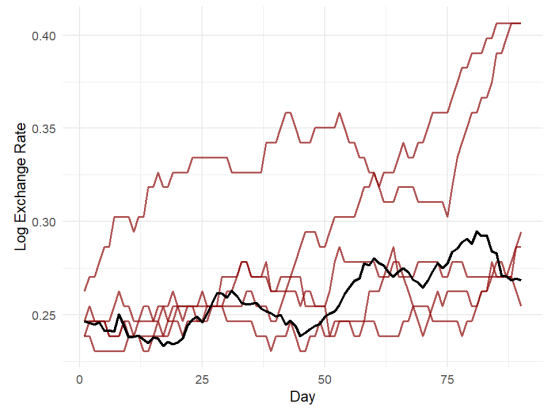
The distribution is symmetric around zero, with most probability concentrated at $\epsilon_t = 0 \approx 58\%$, indicating that exchange rates usually show little movement. Smaller daily changes occur at $\epsilon_t = \pm 1 \approx 18\%$, while larger shocks at $\epsilon_t = \pm 2$ are rare $\approx 2\%$. This reflects the typical stability of liquid currency markets, punctuated by occasional moderate or extreme fluctuations.

Figure 6.2 Interpretation

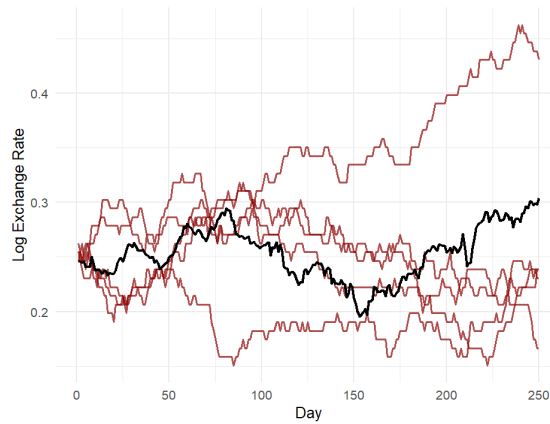
- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Dark red lines: 5 Simulated forecast paths generated by the Random Walk model.
- Black line: Actual test data path used for validation.



(a) Short-term: 3 days



(b) Mid-term: 90 days



(c) Long-term: 250 days

Figure 6.2: Discrete Random Walk simulated plots for GBP/USD

Horizon	RMSE
Short-term (3 days)	0.13
Mid-term (90 days)	1.98
Long-term (250 days)	2.44

Table 6.2: RMSE of Random Walk simulated paths for GBP/USD

6.1.2 Model Fitting for EUR/USD

1. Choice of Scaling Constant K

To discretize the innovations ϵ_t , a scaling constant K was chosen by testing a range of values. For the EUR/USD series, a final choice of $K = 150$ was made, which provided a balance between resolution (capturing finer fluctuations) and concentration (ensuring most of the probability mass was contained within the support).

2. Raw Innovation Distribution ϵ_t

For EUR/USD, most of the scaled innovation values ϵ_t fell between -2 and $+2$, with very little probability mass lying outside this range (e.g., occasional values such as -4 , -3 , 3 , or 5). Since such extreme movements were rare, the distribution was truncated to the support $\{-2, -1, 0, 1, 2\}$ and all probability mass outside this range was proportionally redistributed into the tails.

3. Final Discrete Probability Distribution

After redistributing the mass, the final estimated probabilities of ϵ_t for EUR/USD were:

ϵ_t	Probability (%)
-2	2.18
-1	18.40
0	58.83
1	18.40
2	2.18

Table 6.3: Discrete probability distribution of ϵ_t for EUR/USD.

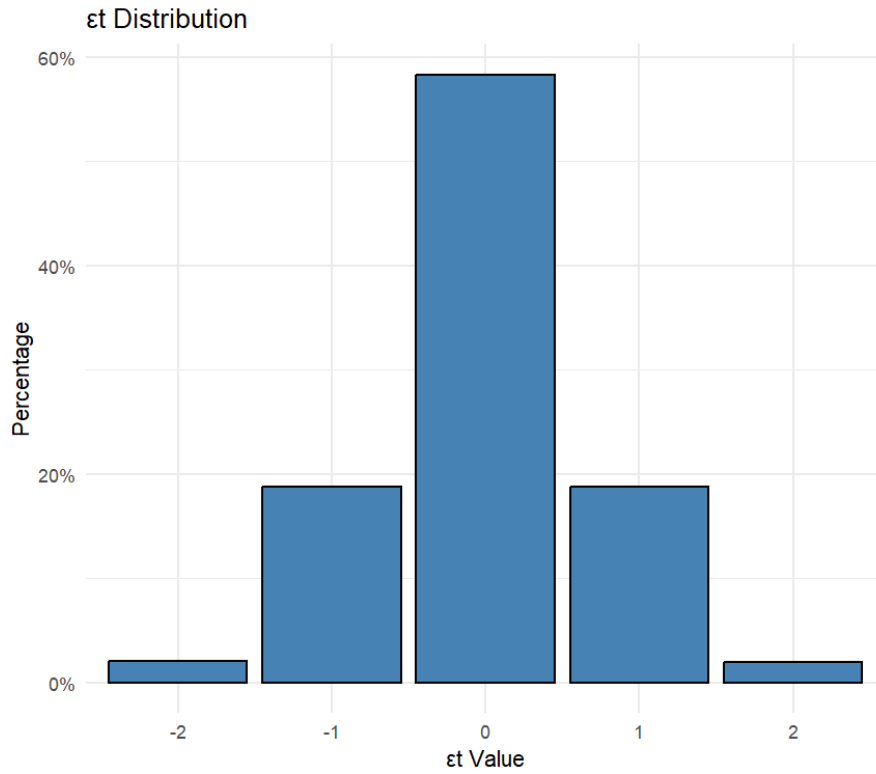


Figure 6.3: Distribution Plot for EUR/USD

The distribution is symmetric around zero, with most probability concentrated at $\epsilon_t = 0 \approx 59\%$, indicating that exchange rates usually show little movement. Smaller daily changes occur at $\epsilon_t = \pm 1 \approx 18\%$, while larger shocks at $\epsilon_t = \pm 2$ are rare $\approx 2\%$. This reflects the typical stability of liquid currency markets, punctuated by occasional moderate or extreme fluctuations.

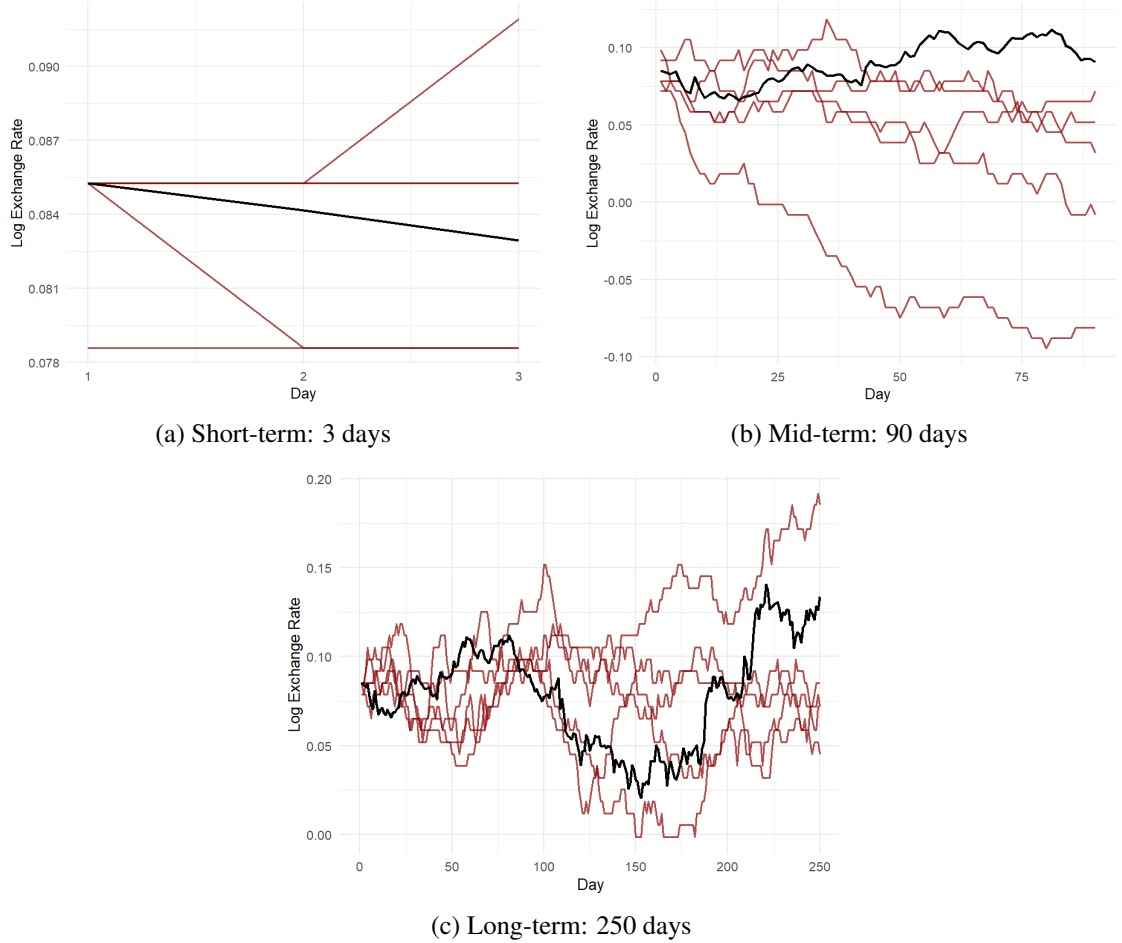


Figure 6.4: Discrete Random Walk simulated paths for EUR/USD

Figure 6.4 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Dark red lines: 5 Simulated forecast paths generated by the Random walk model.
- Black line: Actual test data path used for validation.

Horizon	RMSE
Short-term (3 days)	0.13
Mid-term (90 days)	1.42
Long-term (250 days)	2.92

Table 6.4: RMSE of Random Walk simulated paths for EUR/USD

6.2 ARIMA

To identify the most appropriate time series models for the log exchange rates, the `auto.arima` function in R was applied to both GBP/USD and EUR/USD. This procedure systematically searches through possible ARIMA specifications, selecting the model that best balances goodness-of-fit and parsimony based on information criteria (AIC, BIC). The results reveal that both exchange rates follow dynamics consistent with the random walk hypothesis, with GBP/USD requiring a slight moving average adjustment while EUR/USD is best captured by a pure random walk. The selected models, their estimated parameters, fit statistics, and forecast accuracy measures are summarized below.

6.2.1 ARIMA Model for GBP/USD

Selected Model: ARIMA(0,1,1)

The selected model is an ARIMA(0,1,1). This implies that the series is differenced once ($d = 1$) to achieve stationarity. The specification includes one moving average term (MA(1)) but no autoregressive term.

Coefficient:

- MA(1) = 0.0392 (s.e. = 0.0212)

This indicates a very small but statistically borderline significant short-term dependence in shocks.

Model fit statistics:

- Residual variance: $\sigma^2 = 3.71 \times 10^{-5}$
- Log-likelihood = 8226.19

Information criteria:

- AIC = -16448.38
- BIC = -16436.96

Training set accuracy:

- RMSE = 0.0061
- MAE = 0.0043
- Mean percentage error (MPE) $\approx -0.066\%$, showing no systematic bias
- Mean absolute percentage error (MAPE) $\approx 1.79\%$
- MASE ≈ 1.00 , suggesting performance comparable to a random walk
- ACF1 ≈ 0 , indicating little residual autocorrelation

ACF and PACF Plots

This figure shows the ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) plots of the residuals from the fitted ARIMA(0,1,1) model.

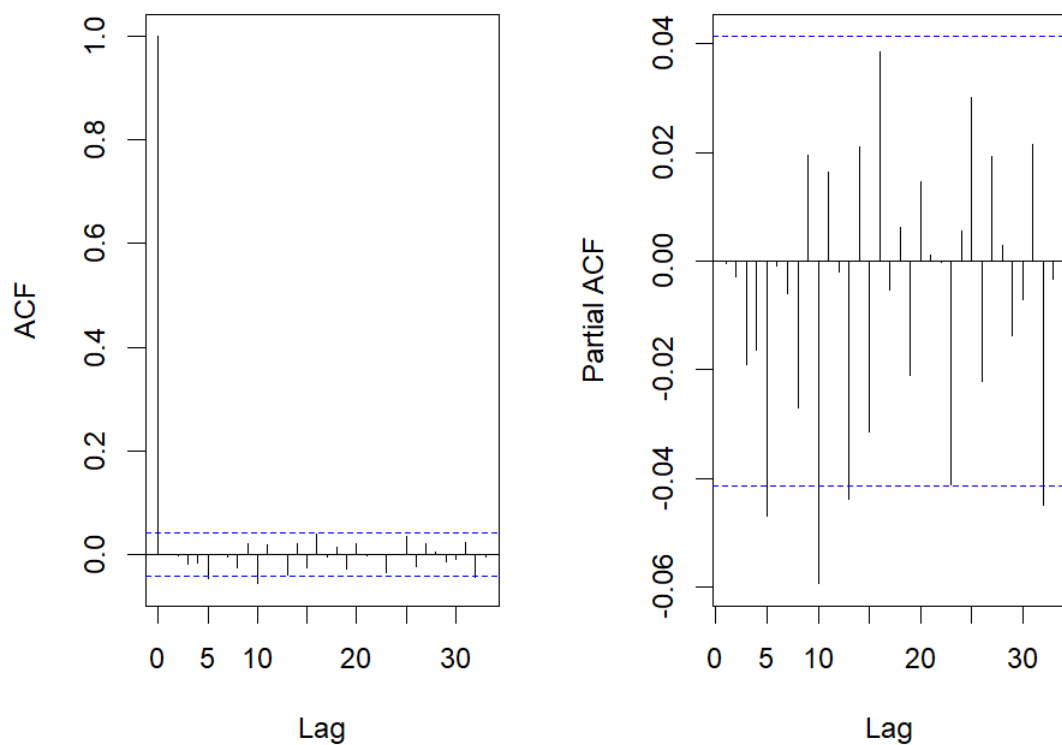


Figure 6.5: ACF and PACF Plots for GBP/USD

Left: ACF plot

- The large spike at lag 1 is due to the first differencing ($d=1$), which removes the trend.
- After lag 1, the autocorrelations quickly fall within the confidence bounds (blue dashed lines), suggesting that the residuals behave like white noise.
- This indicates the ARIMA(0,1,1) model has adequately captured the serial dependence in the data.

Right: PACF plot

- The partial autocorrelations fluctuate randomly around zero, with no strong significant spikes beyond the confidence limits.
- This reinforces the idea that no additional AR terms are needed.

The diagnostic plots confirm that the ARIMA(0,1,1) model is appropriate:

- Differencing ($d=1$) made the series stationary.
- The MA(1) component captured the short-term dependence.
- Residuals resemble white noise, meaning the model sufficiently explains the structure in the data.

Figure 6.6 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Red lines: 5 Simulated forecast paths generated by the ARIMA(0,1,1) model.
- Black line: Actual test data path used for validation.

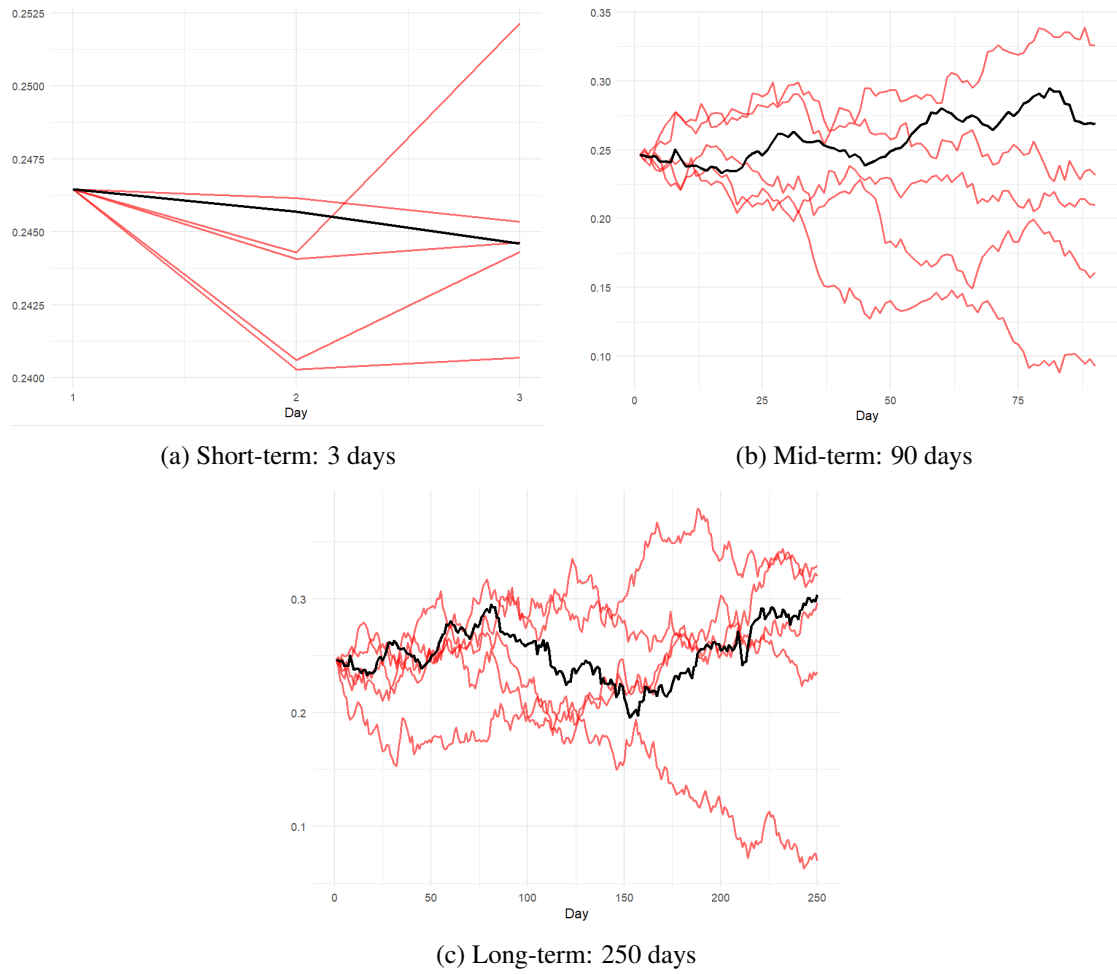


Figure 6.6: ARIMA simulated paths for GBP/USD

RMSE Results

Table 6.5: RMSE of ARIMA(0,1,1) forecasts for GBP/USD

Forecast Horizon	RMSE
3 Days	0.12
90 Days	2.04
250 Days	2.55

6.2.2 ARIMA Model for EUR/USD

Selected Model: ARIMA(0,1,0)

The selected model is an **ARIMA(0,1,0)**, which corresponds to a pure random walk. The series is differenced once ($d = 1$) to achieve stationarity, with no autoregressive (AR) or moving average (MA) components.

Model fit statistics:

- Residual variance: $\sigma^2 = 2.34 \times 10^{-5}$
- Log-likelihood = 8742.69

Information criteria:

- AIC = -17483.37
- BIC = -17477.66

Training set accuracy:

- RMSE = 0.0048
- MAE = 0.0036
- MASE ≈ 1.00 , consistent with a random walk
- ACF1 ≈ 0 , indicating residuals behave like white noise
- MPE and MAPE are reported as ∞ due to near-zero denominators in percentage error calculation

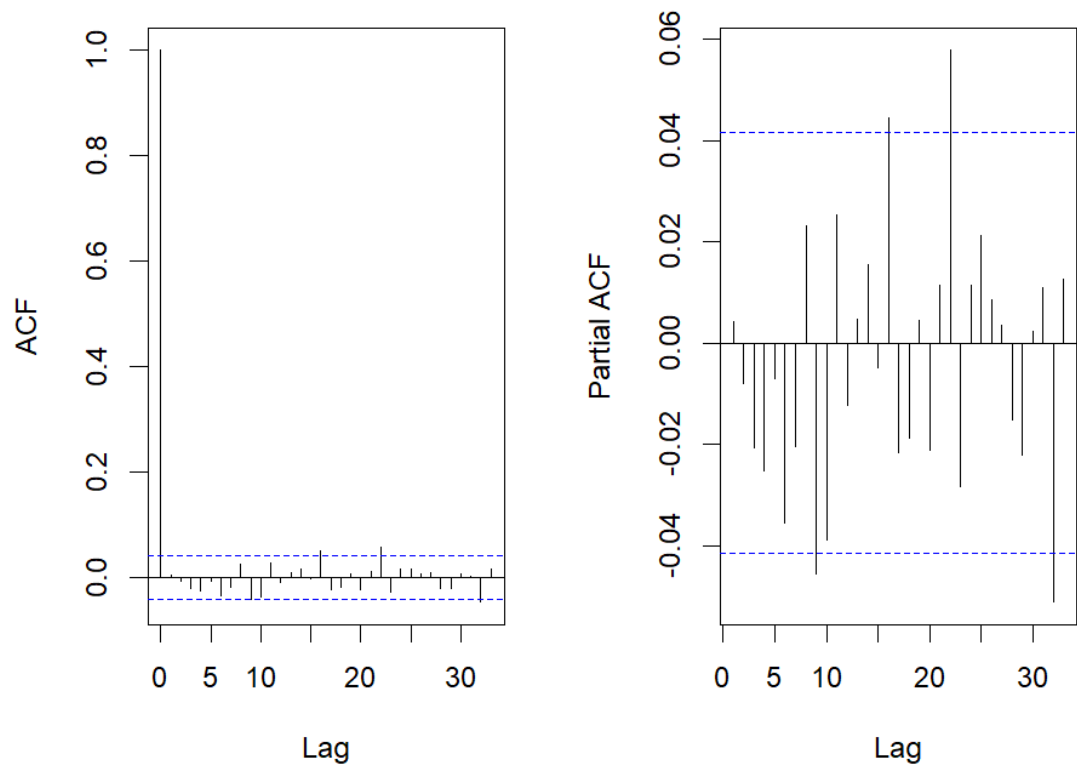


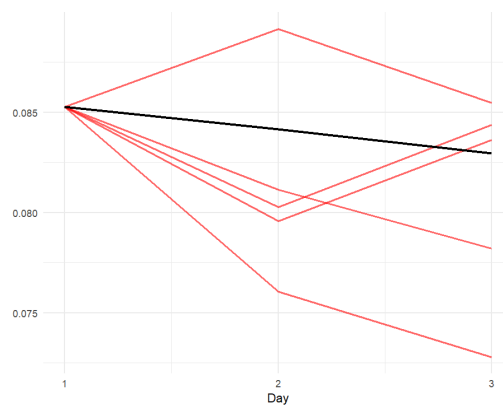
Figure 6.7: ACF and PACF Plots for EUR/USD

Left: ACF plot

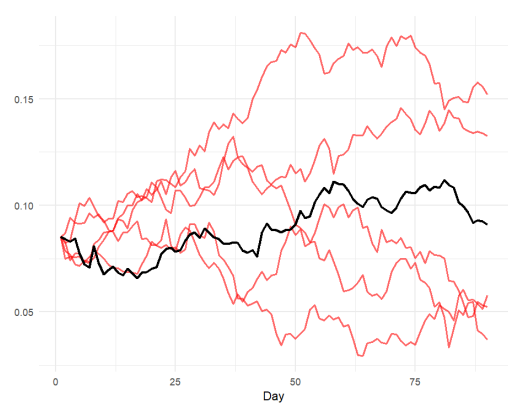
- The autocorrelations drop to near zero immediately after lag 0.
- This is characteristic of a random walk (ARIMA(0,1,0)) model, where the differenced series has no significant autocorrelation.

Right plot (PACF):

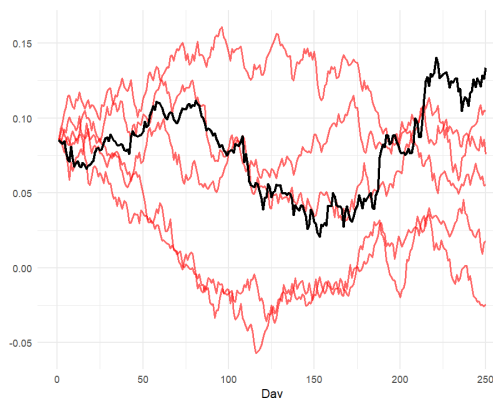
- Partial autocorrelations also fluctuate around zero and do not show significant spikes beyond lag 0.
- This further confirms that there is no AR structure in the differenced series.
- ARIMA(0,1,0) means: no autoregressive terms (AR=0), differencing once (I=1), and no moving average terms (MA=0).
- Essentially, the original series follows a random walk, and the first difference of the series behaves like white noise (uncorrelated).



(a) Short-term: 3 days



(b) Mid-term: 90 days



(c) Long-term: 250 days

Figure 6.8: ARIMA simulated paths for EUR/USD

Figure 6.8 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Red lines: 5 Simulated forecast paths generated by the ARIMA(0,1,0) model.
- Black line: Actual test data path used for validation.

RMSE Results

Table 6.6: RMSE of ARIMA(0,1,0) forecasts for EUR/USD

Forecast Horizon	RMSE
3 Days	0.14
90 Days	1.42
250 Days	3.04

6.3 White Noise Random walk model

In this study, the WNRW model was applied to two major currency pairs, GBP/USD and EUR/USD, which are among the most actively traded exchange rates. By estimating the drift and volatility parameters from historical log-returns of each series, the model provides insights into their average behavior and variability. Simulated paths based on these parameters allow for the assessment of short-, mid-, and long-term forecast dynamics, offering a benchmark against which more complex models can be compared.

6.3.1 White Noise Model For GBP/USD

The log-returns of the GBP/USD training data were computed as

$$r_t = \log(S_t) - \log(S_{t-1}).$$

The estimated parameters are:

$$\hat{\mu} = -9.71 \times 10^{-5}, \quad \hat{\sigma} = 0.00609.$$

Interpretation: The drift is very close to zero, suggesting that GBP/USD exhibits no strong upward or downward trend on average. The volatility ($\approx 0.61\%$) represents the typical daily fluctuation in log exchange rates.

Parameter	Estimate
Drift ($\hat{\mu}$)	-9.71×10^{-5}
Volatility ($\hat{\sigma}$)	0.00609

Table 6.7: Estimated drift and volatility for GBP/USD.

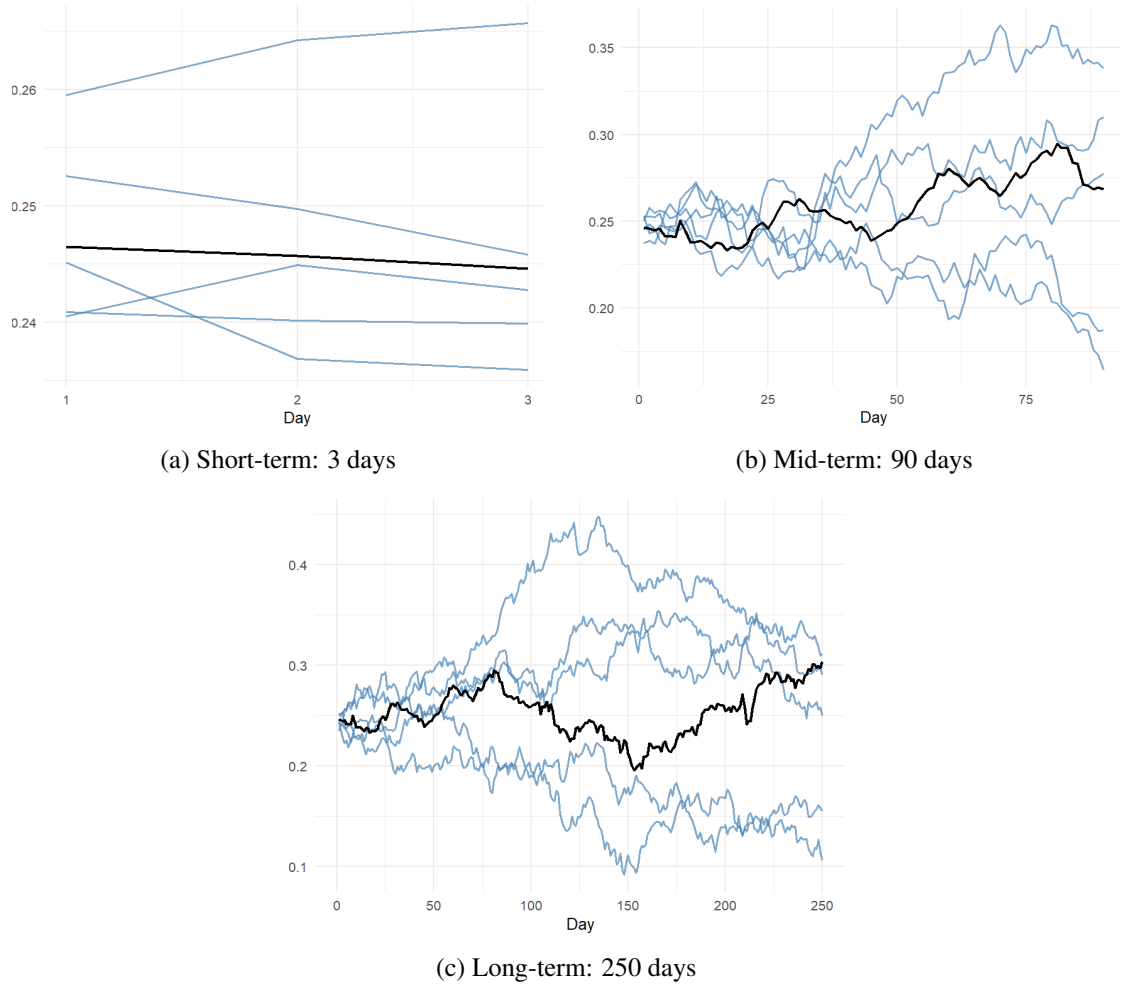


Figure 6.9: White Noise Random Walk simulated paths for GBP/USD

Figure 6.9 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Blue lines: 5 Simulated forecast paths generated by the White Noise model.
- Black line: Actual test data path used for validation.

RMSE Results

Horizon	RMSE
Short-term (3 days)	0.12
Mid-term (90 days)	2.60
Long-term (250 days)	3.22

Table 6.8: RMSE of White Noise Random Walk for GBP/USD

6.3.2 White Noise Model for EUR/USD

The log-returns of the EUR/USD training data were computed as

$$r_t = \log(S_t) - \log(S_{t-1}).$$

The estimated parameters are:

$$\hat{\mu} = -2.14 \times 10^{-5}, \quad \hat{\sigma} = 0.00483.$$

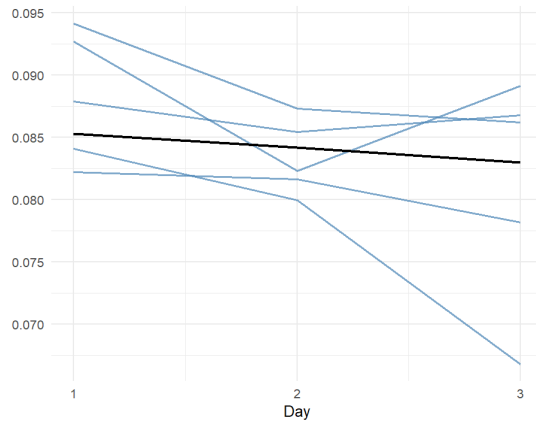
Interpretation: Similar to GBP/USD, the drift is near zero, indicating little directional bias. The volatility is slightly lower than GBP/USD ($\approx 0.48\%$), suggesting relatively smaller daily fluctuations.

Parameter	Estimate
Drift ($\hat{\mu}$)	-2.14×10^{-5}
Volatility ($\hat{\sigma}$)	0.00483

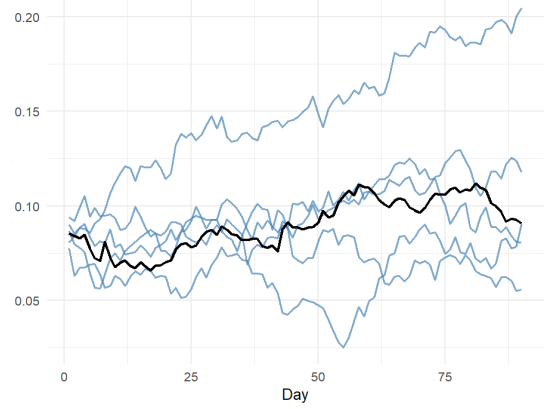
Table 6.9: Estimated drift and volatility for EUR/USD.

Figure 6.10 Interpretation

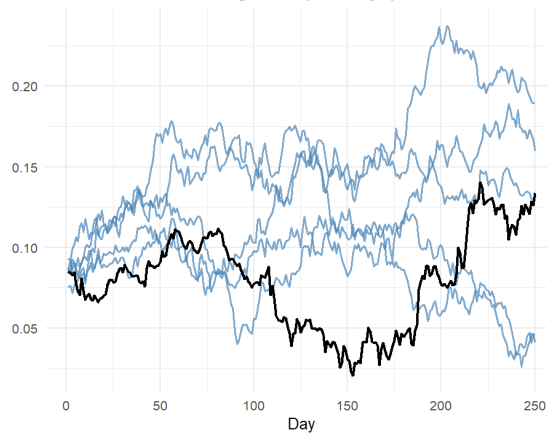
- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Blue lines: 5 Simulated forecast paths generated by the White Noise model.
- Black line: Actual test data path used for validation.



(a) Short-term: 3 days



(b) Mid-term: 90 days



(c) Long-term: 250 days

Figure 6.10: White Noise Random Walk simulated paths for EUR/USD

RMSE Results

Horizon	RMSE
Short-term (3 days)	0.14
Mid-term (90 days)	1.61
Long-term (250 days)	2.95

Table 6.10: RMSE of White Noise Random Walk for EUR/USD

6.4 GARCH Model

In this study, we estimate and evaluate GARCH(1,1) models for two of the world's most actively traded currency pairs—GBP/USD and EUR/USD. By analyzing the behavior of their returns and volatility, the models provide insights into how shocks impact market fluctuations and whether volatility patterns are consistent across different exchange rates.

The GARCH(1,1) model was selected because it strikes a balance between parsimony and effectiveness in modeling financial time series. Exchange rate returns typically exhibit volatility clustering (periods of high volatility followed by calm periods) and persistent conditional variance, both of which the GARCH(1,1) can capture with just two parameters:

- α_1 (ARCH effect) measures how volatility reacts to new shocks.
- β_1 (GARCH effect) captures the persistence of volatility over time.

6.4.1 GARCH Model for GBP/USD

1. Model Estimation Results

- Estimated mean return:

$$\mu = -0.000059$$

- The coefficient is very small and statistically insignificant ($p = 0.51$).
- Interpretation: The average daily log-return of GBP/USD is essentially zero, consistent with the Efficient Market Hypothesis (no predictable drift in returns).

2. Variance Equation

- Constant term: $\omega \approx 0$. The unconditional baseline variance is negligible.
- ARCH effect: $\alpha_1 = 0.042$ (significant). About 4.2% shocks feed into volatility, but the impact is moderate.
- GARCH effect: $\beta_1 = 0.950$ (highly significant). Volatility is extremely persistent, with $\sim 95\%$ of past variance carried forward.
- Persistence: $\alpha_1 + \beta_1 \approx 0.992$, very close to 1, indicating long memory in volatility and the presence of volatility clustering (large shocks are followed by large shocks).

3. Model Fit

- Log-likelihood: 8868.61
- AIC: -7.936 (favorable compared to alternative specifications).
- Interpretation: The GARCH(1,1) is a well-fitting and parsimonious model for GBP/USD log-returns.

4. **Overall Interpretation** The GBP/USD market shows:

- No significant drift (returns ≈ 0 on average).
- Volatility that is highly persistent but reacts moderately to new shocks.
- Well-captured volatility clustering, consistent with typical FX market behavior.

Diagnostic Checks

1. **Ljung–Box Test**

H_0 : No serial correlation, H_1 : Serial correlation is present.

The test produced a p -value of 0.5445. Since $p > 0.05$, we fail to reject H_0 , concluding that no significant autocorrelation remains in the residuals.

2. **ARCH LM Test**

H_0 : No ARCH effects, H_1 : Presence of ARCH effects.

The test returned a p -value of 0.6178. Again, $p > 0.05$ implies failure to reject H_0 , showing that volatility clustering is adequately captured.

Since both tests confirm that the residuals are well behaved (i.e., white noise with no remaining ARCH effects), there is no need to consider more complex GARCH specifications. Therefore, the GARCH(1,1) model was retained as the final model.

Figure 6.11 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Orange lines: 5 Simulated forecast paths generated by the GARCH model.
- Black line: Actual test data path used for validation.

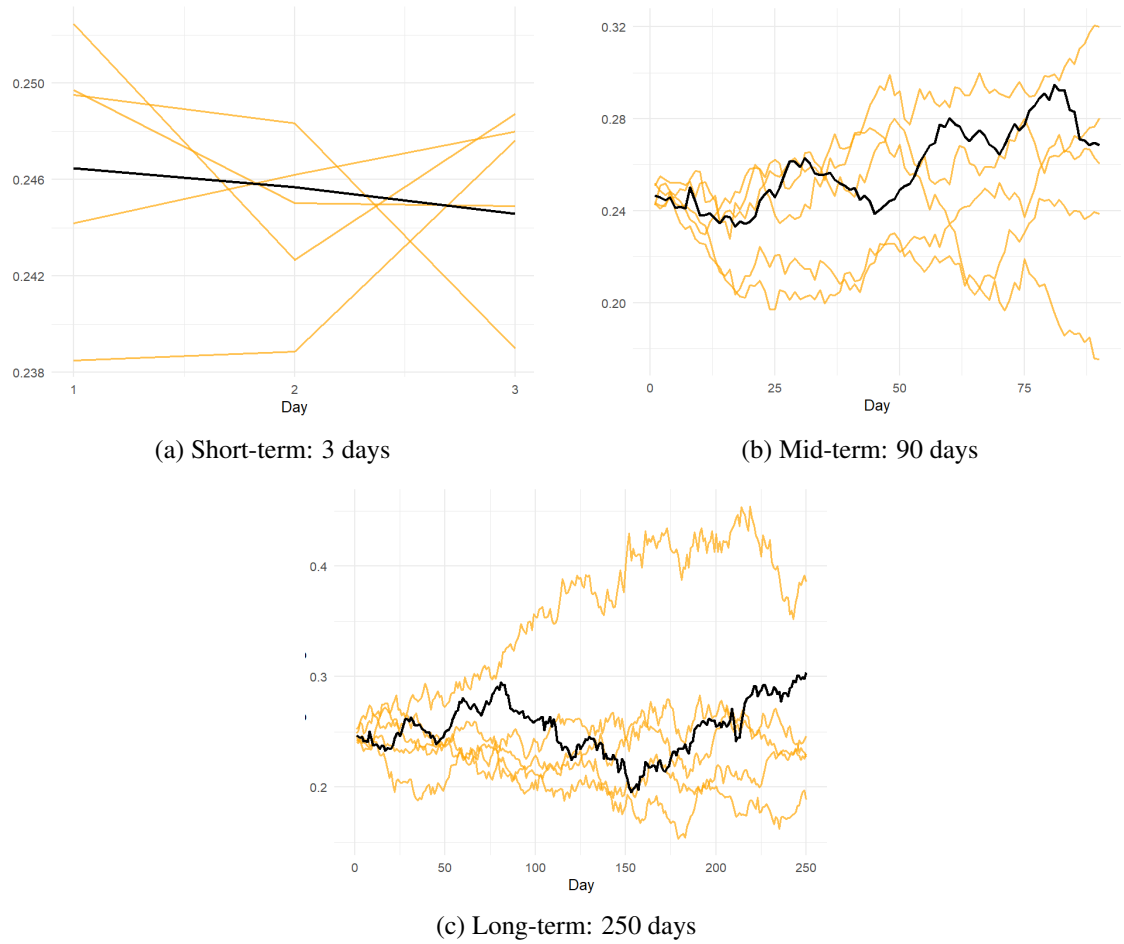


Figure 6.11: GARCH simulated paths for GBP/USD

RMSE Result

Table 6.11: Forecast RMSE of GARCH(1,1) Model for GBP/USD

Forecast Horizon	RMSE
Short-term (3 days)	0.12
Mid-term (90 days)	2.37
Long-term (250 days)	2.89

6.4.2 GARCH Model for EUR/USD

1. Mean Equation

- Estimated mean return:

$$\mu = -0.000059$$

- The coefficient is not statistically significant ($p \approx 0.74$, with robust standard errors).

- Interpretation: The average daily log-return of EUR/USD is essentially zero, consistent with the Efficient Market Hypothesis (no predictable drift in returns).

2. Variance Equation

- Constant term: $\omega \approx 0$. The unconditional baseline variance is negligible.
- ARCH effect: $\alpha_1 = 0.042$ (significant). Moderate response of volatility to new shocks.
- GARCH effect: $\beta_1 = 0.950$ (highly significant). Strong persistence, with volatility largely carried over time.
- Persistence: $\alpha_1 + \beta_1 \approx 0.992$, suggesting long memory in volatility and the presence of volatility clustering.

3. Model Fit

- Log-likelihood: 8868.61
- AIC: -7.936 (same as GBP/USD, suggesting identical fit quality).
- Interpretation: The GARCH(1,1) provides a parsimonious and robust fit for EUR/USD log-returns.

4. Overall Interpretation The EUR/USD market exhibits:

- No significant drift (average return ≈ 0).
- Persistent and clustered volatility, reacting moderately to shocks.
- A symmetric GARCH(1,1) with normal errors adequately explains volatility, mirroring the dynamics observed in GBP/USD.

Diagnostic Checks

1. Ljung–Box Test

H_0 : No serial correlation, H_1 : Serial correlation is present.

The test produced a p -value of 0.5766. Since $p > 0.05$, we fail to reject H_0 , indicating that there is no significant autocorrelation in the residuals.

2. ARCH LM Test

H_0 : No ARCH effects, H_1 : Presence of ARCH effects.

The test yielded a p -value of 0.8839. Since $p > 0.05$, we fail to reject H_0 , confirming that the conditional heteroskedasticity has been adequately captured by the GARCH(1,1) model.

The test yielded a p -value of 0.8839. Since $p > 0.05$, we fail to reject H_0 , confirming that conditional heteroskedasticity is well modeled.

Because both tests show that the standardized residuals are free from autocorrelation and additional ARCH effects, the GARCH(1,1) specification is sufficient. Adding more lags or moving to a higher-order GARCH model would not significantly improve model fit. Thus, we stopped at GARCH(1,1) since it parsimoniously explains the volatility clustering without overfitting.

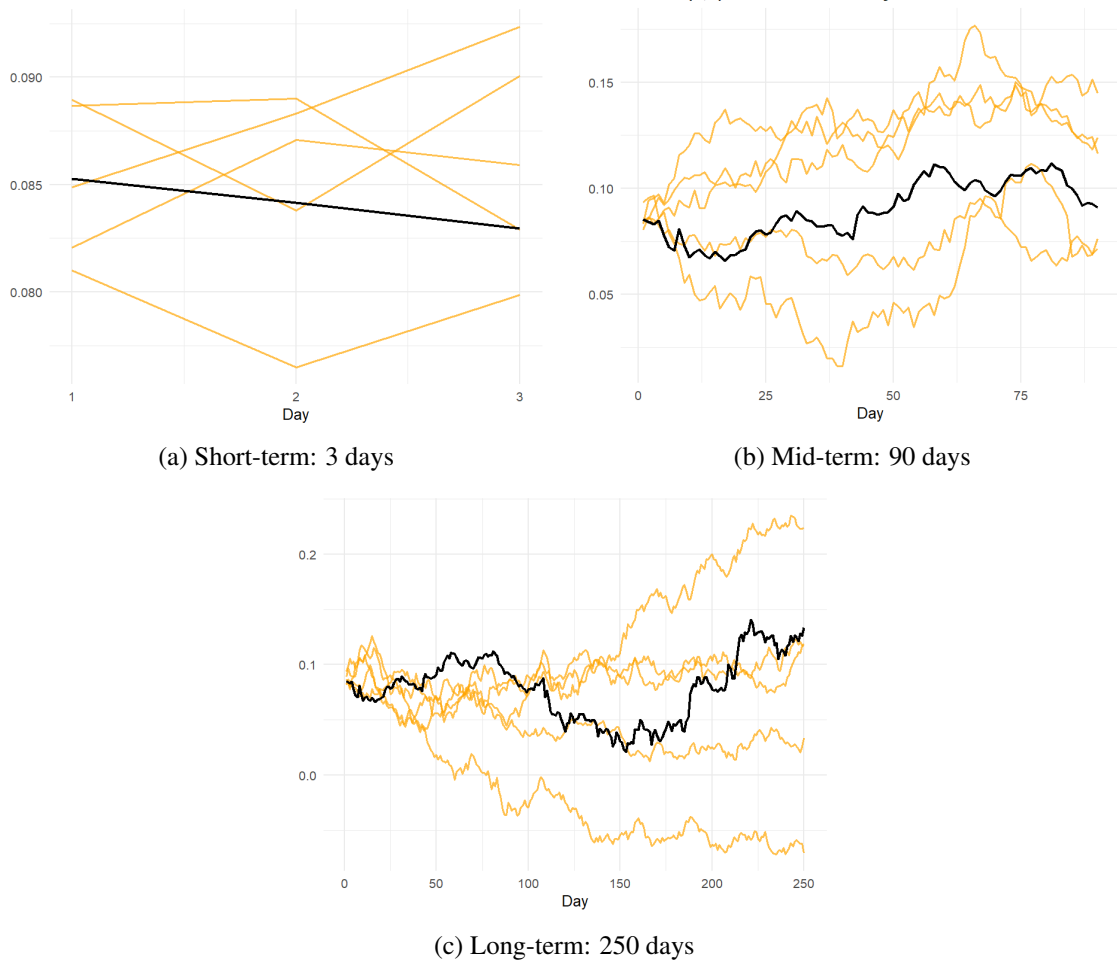


Figure 6.12: GARCH simulated paths for EUR/USD

Figure 6.12 Interpretation

- X-axis (Day): Represents the forecast horizon (3, 90, and 250 days in the three panels).
- Y-axis: Represents forecast errors or forecasted exchange rate levels.
- Orange lines: 5 Simulated forecast paths generated by the GARCH model.
- Black line: Actual test data path used for validation.

RMSE Result

Table 6.12: Forecast RMSE of GARCH(1,1) Model for EUR/USD

Forecast Horizon	RMSE
Short-term (3 days)	0.12
Mid-term (90 days)	1.50
Long-term (250 days)	3.06

6.5 Summary

This section compares the forecast accuracy of GBP/USD and EUR/USD across different models and horizons using RMSE. It highlights how simple models like Random Walk and ARIMA perform well in the short term, while volatility-aware models such as GARCH provide modest improvements for medium-term forecasts. The analysis emphasizes the increasing uncertainty of long-term exchange rate predictions.

GBP/USD RMSE Results

Table 6.13: RMSE Comparison of GBP/USD Forecasts Across Models and Horizons

Model	Short-term (3 days)	Mid-term (90 days)	Long-term (250 days)
Random Walk	0.13	1.98	2.44
ARIMA(0,1,1)	0.12	2.04	2.55
White Noise	0.12	2.60	3.22
GARCH(1,1)	0.12	2.37	2.89

- **Short-term forecasts (3 days):** RMSE values are very low (0.12–0.13%), showing that near-term exchange rate movements are predictable regardless of model complexity.
- **Medium-term forecasts (90 days):** Errors increase as the horizon extends. On the log scale, volatility-aware models like GARCH (2.38%) slightly outperform the White Noise (2.60%) in capturing relative fluctuations.
- **Long-term forecasts (250 days):** Substantial error accumulation is observed (absolute RMSE 2.44 - 2.90%, $\text{RMSE} \approx 3\%$), underlining the inherent uncertainty in forecasting exact exchange rate levels over long horizons.

The Random Walk model consistently outperforms the other forecasting models across all horizons. It achieves the near lower RMSE for short-term (3 days) and lowest RMSE for mid-term (90 days), and long-term (250 days) forecasts, suggesting that, despite its simplicity, the Random Walk provides more accurate predictions for GBP/USD than ARIMA(0,1,1), White Noise, or GARCH(1,1) models.

EUR/USD RMSE Results

Table 6.14: RMSE Comparison of EUR/USD Forecasts Across Models and Horizons

Model	Short-term (3 days)	Mid-term (90 days)	Long-term (250 days)
Random Walk	0.13	1.42	2.92
ARIMA(0,1,0)	0.14	1.42	3.04
White Noise	0.14	1.61	2.95
GARCH(1,1)	0.12	1.50	3.06

- **Short-term forecasts (3 days):** RMSE values are very low (0.12 - 0.14%), showing that near-term exchange rate movements are highly predictable and relatively insensitive to model complexity.
- **Medium-term forecasts (90 days):** Errors increase slightly. On the log scale, volatility-aware models like GARCH (1.51%) perform comparably to Random Walk ARIMA (1.42%) and slightly better than the White Noise (1.61%) in capturing relative fluctuations.
- **Long-term forecasts (250 days):** Substantial error accumulation is observed (absolute RMSE 2.92 - 3.06%, $\text{RMSE} \approx 2.95\%$), highlighting the inherent uncertainty in predicting exact exchange rate levels over longer horizons.

The Random Walk model demonstrates better overall performance compared to the alternative models. It achieves the lowest or near-lowest RMSE in short-term (3 days) and mid-term (90 days) forecasts and maintains competitive accuracy in the long-term (250 days), indicating that, despite its simplicity, the Random Walk provides more reliable predictions than ARIMA(0,1,0), White Noise, or GARCH(1,1) models.

Chapter 7

Conclusion

The analysis of GBP/USD and EUR/USD forecasts across multiple models and horizons reveals key insights about exchange rate predictability. Short-term forecasts are relatively predictable and show little sensitivity to model complexity, whereas medium-term forecasts benefit modestly from volatility-aware models that capture fluctuations more effectively. Long-term forecasts, however, remain highly uncertain, reflecting the inherent difficulty of accurately predicting exchange rate levels over extended periods. Across both currency pairs, the Random Walk model consistently demonstrates robust performance compared to more complex alternatives. This suggests that, despite the sophistication of models like ARIMA and GARCH, simple persistence-based approaches provide a reliable benchmark for forecasting and are often as effective, if not more, than advanced models—particularly when accounting for the high uncertainty of longer horizons. These findings highlight the value of simplicity in foreign exchange forecasting and the limited incremental benefit of model complexity for practical prediction purposes.

Overall, this research confirms the robustness of random walk models as a forecasting benchmark, while showing how discretized innovations can extend their utility in practice. Future work may explore hybrid approaches that combine this method with machine learning or structural macroeconomic models to enhance predictive power, particularly in volatile or crisis-prone environments.

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